Lab 5

# Tracking Organ Donations

KDE Plots, Logistic Regression, Census Regions & Divisions

**DATA 20195 Spring 2026**

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# Organ Procurement and Transplantation Network (OPTN)

- To address the nation's critical organ donation shortage and improve the **organ matching** and placement process, the **U.S. Congress passed the National Organ Transplant Act in 1984**:
  - established the OPTN to maintain a national registry for organ matching
  - declared the network should be operated by a private organization under federal contract

- OPTN leads the network of transplant hospitals, organ procurement organizations, volunteers

## Illinois waiting list

| | All Organs | Kidney | Liver | Pancreas | Kidney / Pancreas | Heart | Lung | Intestine |
|---|---|---|---|---|---|---|---|---|
| Candidates | 4,432 | 3,855 | 203 | 80 | 158 | 207 | 46 | 10 |

## U.S. waiting list

| | All Organs | Kidney | Liver | Pancreas | Kidney / Pancreas | Heart | Lung | Heart / Lung | Intestine | VCA - abdominal wall | VCA - head and neck | VCA - other genitourinary organ | VCA - upper limb | VCA - uterus |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Candidates | 109,126 | 94,818 | 8,868 | 828 | 2,389 | 4,063 | 1,023 | 30 | 169 | 1 | 2 | 2 | 2 | 3 |

General info: https://www.hrsa.gov/optn/about
Data source: https://hrsa.unos.org/data/view-data-reports/state-data/?type=state&selectAreaValue=IL#, https://hrsa.unos.org/data/view-data-reports/national-data/#
Multiple listings info: https://www.hrsa.gov/optn/patients/organ-transplants/multiple-listing

# STAR (Standard Transplant Analysis and Research) file

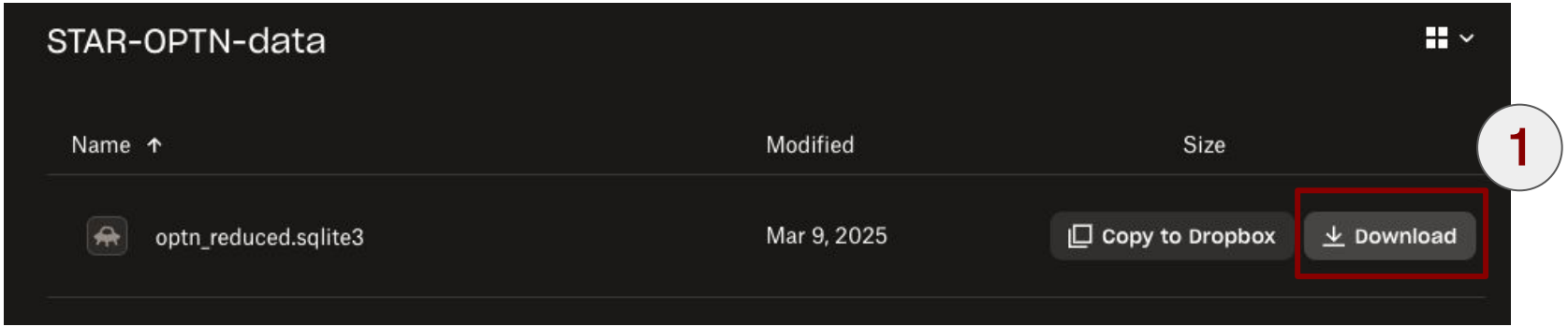
- OPTN releases data in **STAR (Standard Transplant Analysis and Research)** files
  - limited datasets that contain patient–level information about transplant recipients, deceased and living donors, and waiting list candidates back to 10/1/1987
  - updated quarterly
  - all STAR file requests require a signed data use agreement and are delivered via a secure electronic format

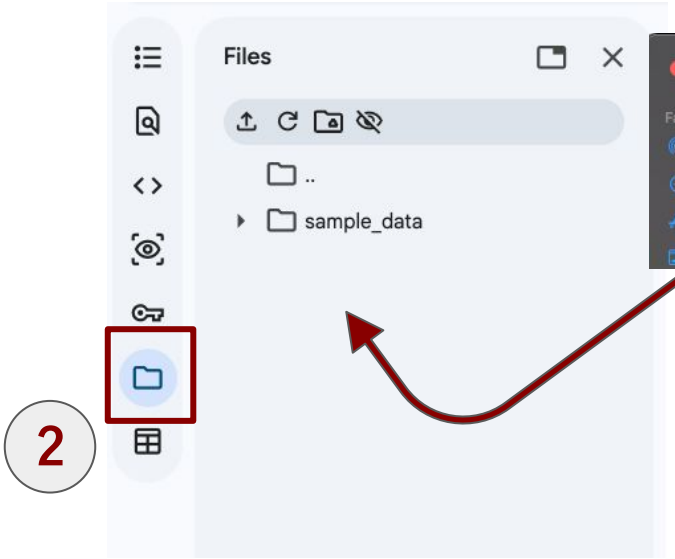
| | |
|---|---|
| PCO2_DON | DDR ABG RESULTS: PCO2: |
| PCWP_INIT_DON | DDR PCWP PA CATHETER INITIAL (MM/HG) |
| PCWP_POST_DON | DDR PCWP PA CATHETER FINAL (MM/HG) |
| PH_DON | DDR ABG RESULTS: BLOOD PH: |
| PHSBORNTOMOTHERWITHHIVHBVHCV | Child born to a mother with HIV, HBV, or HCV infection |
| PHSBREASTFEDBYMOTHERWITHHIV | Child breastfed by a mother with HIV infection |
| PHSDRUGINJECTION | Drug injection for nonmedical reasons |
| PHSINCARCERATION72 | Incarceration (confinement in jail, prison, or juvenile correction facility) for 72 or more consecutive hours |
| PHSSEXUALCONTACTDRUGINJECTION | Sex with a person who injected drugs for nonmedical reasons |
| PHSSEXUALCONTACTHIVHBVHCV | Sex (i.e., any method of sexual contact, including vaginal, anal, and oral) with a person known or suspected to have HIV, HBV, or HCV infection |
| PHSSEXUALCONTACTMALEMALE | Man who has had sex with another man |
| PHSSEXUALCONTACTMONEY | Sex in exchange for money or drugs |
| PHSSEXUALCONTACTMONEYINDIRECT | Sex with a person who had sex in exchange for money or drugs |
| PHSUNKNOWNMEDSOCIALHISTORY | Unknown medical or social history |
| PO2_DON | DDR ABG RESULTS: PO2 |
| PO2_DONE_DON | DDR:Lung - Was pO2 done: |
| PO2_FIO2_DON | DDR ABG RESULTS: FIO2 |

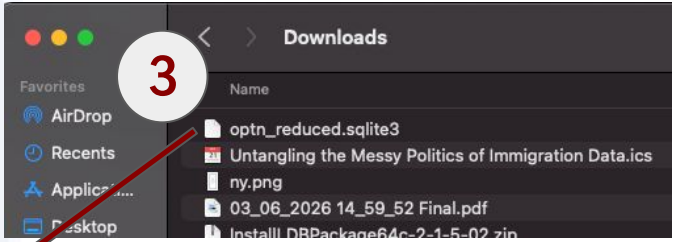
*Examples of information collected about donors. If curious, check out the STAR File Data Dictionary.*

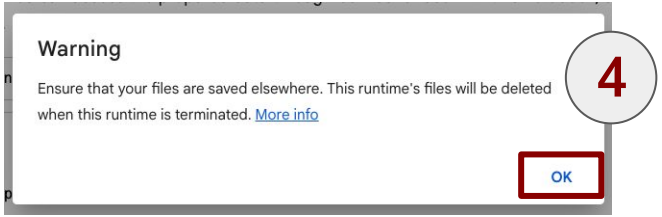
https://hrsa.unos.org/DataRequestInstructions
OPTN STAR File Data Dictionary

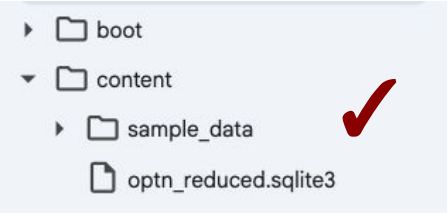
# Downloading & putting dataset into Google Collab:



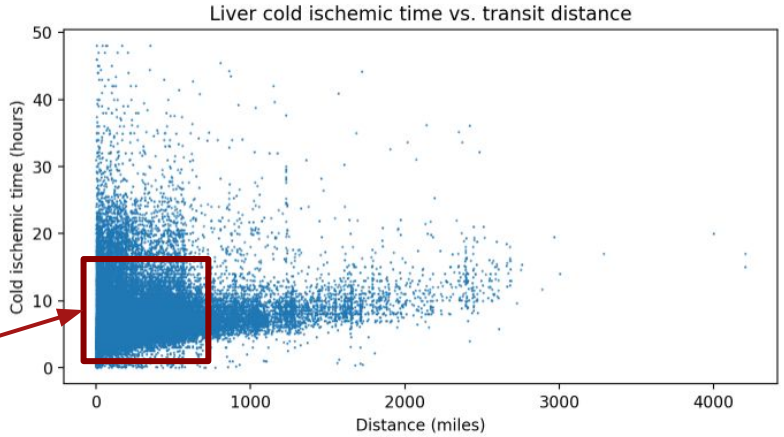



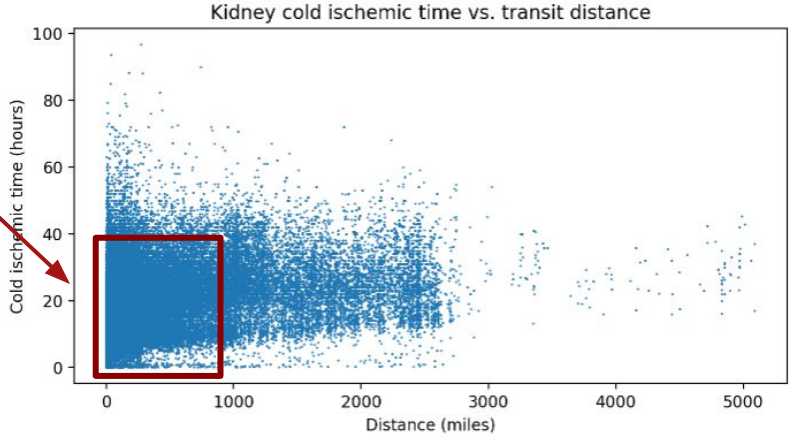



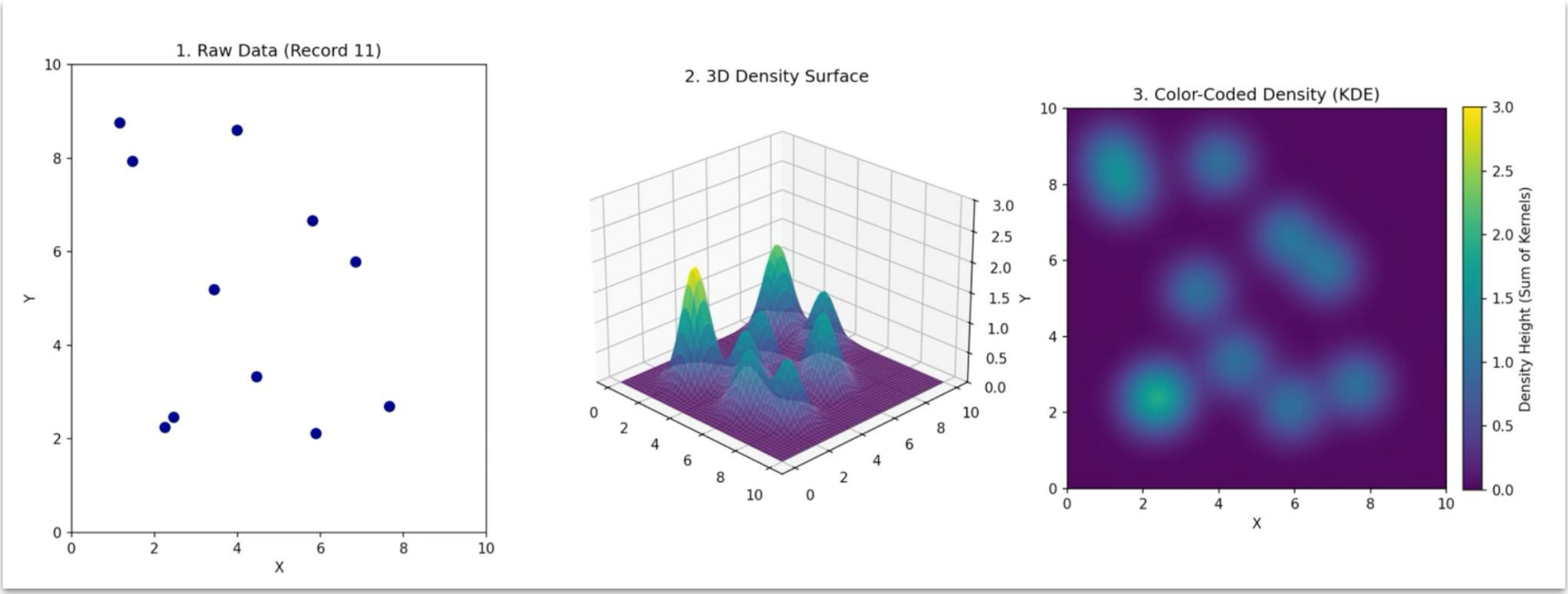
# Discussion – Scatterplot Limitations

*How many points are in these areas?*
*Is it more dense for livers or kidneys?*



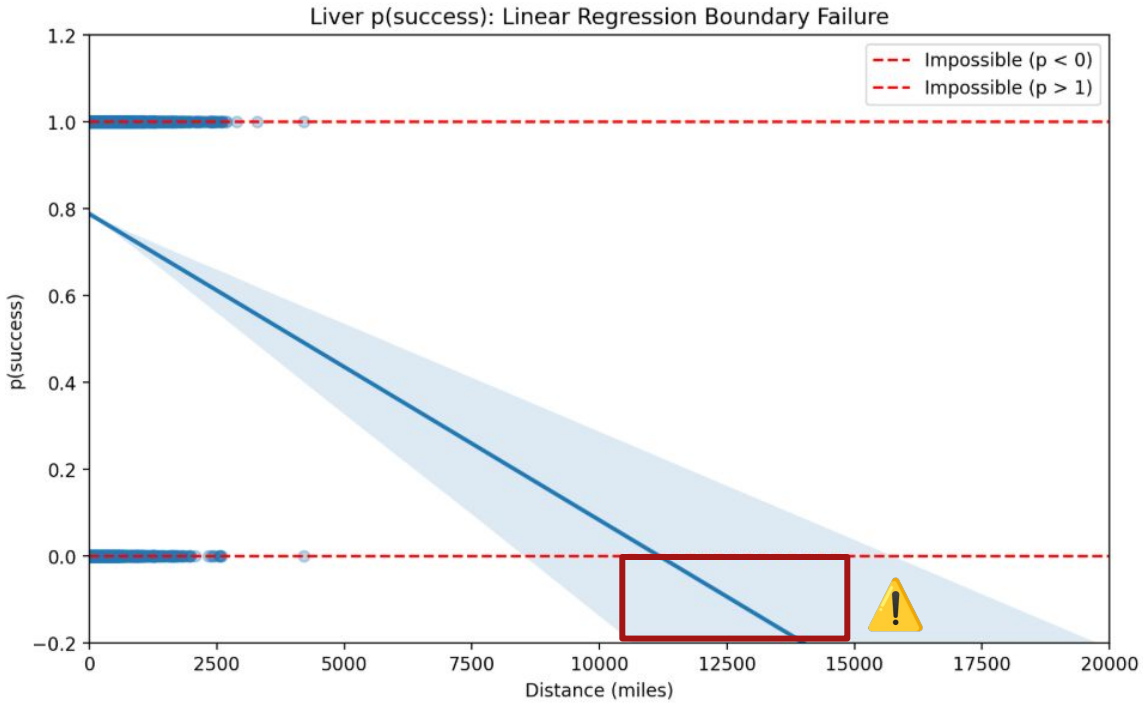
# Kernel Density Estimation




Link to animation:
https://drive.google.com/file/d/1BdmSvykO9rDcRAmVgusewzqrY6lwhwZ1/view?usp=sharing

# Can we use linear regression?

In a standard linear regression, the model predicts the outcome y directly.




But when y is a binary outcome (like an organ transplant succeeding or failing), the output must be a probability, which needs to be bounded between 0 and 1.

# Simple Logistic Regression – Example

Goal: Relate data (e.g. `hours studying`) to a yes/no outcome (e.g. `pass exam`)

Model: `probability of` **`passing exam`** `=` **`f`**`(slope *` **`hours`** `+ intercept)`

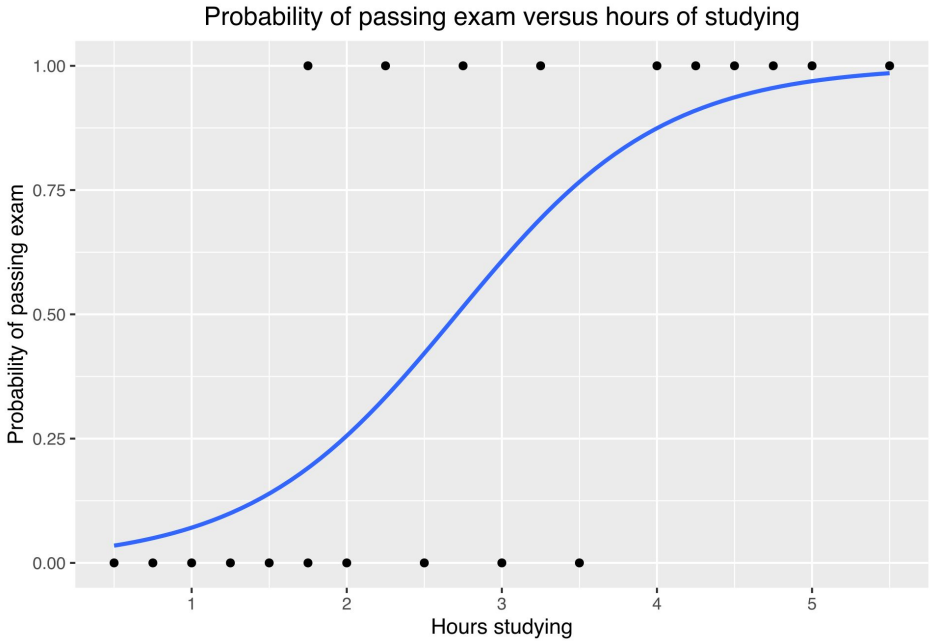



Image credit: Wikipedia

# Simple Logistic Regression

Let probability of passing an exam, p = .75, then odds = .75 / (1 - .75) = .75/ .25 = 3

*for every 3 times you pass, you fail once*

$$\text{Odds} = \frac{p}{1-p}$$

*Odds can range from 0 to positive infinity, but will never be negative, given 0 < p < 1 by definition.*

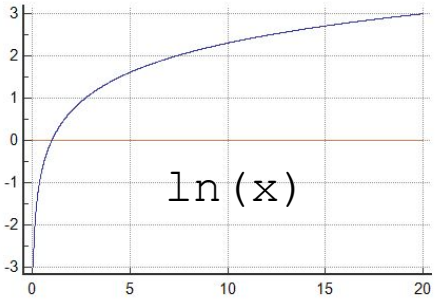
Take the natural log of the odds, which is formally called the 'logit': $\text{Logit}(p) = \ln\left(\frac{p}{1-p}\right)$

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1$$

*Linear relationship between variable(s) and the **Log-Odds**.*

*For every 1 increase in x, the **Log-Odds** of the event succeeding increases by beta.*

We solve the logit equation for p, and this makes the S curve:

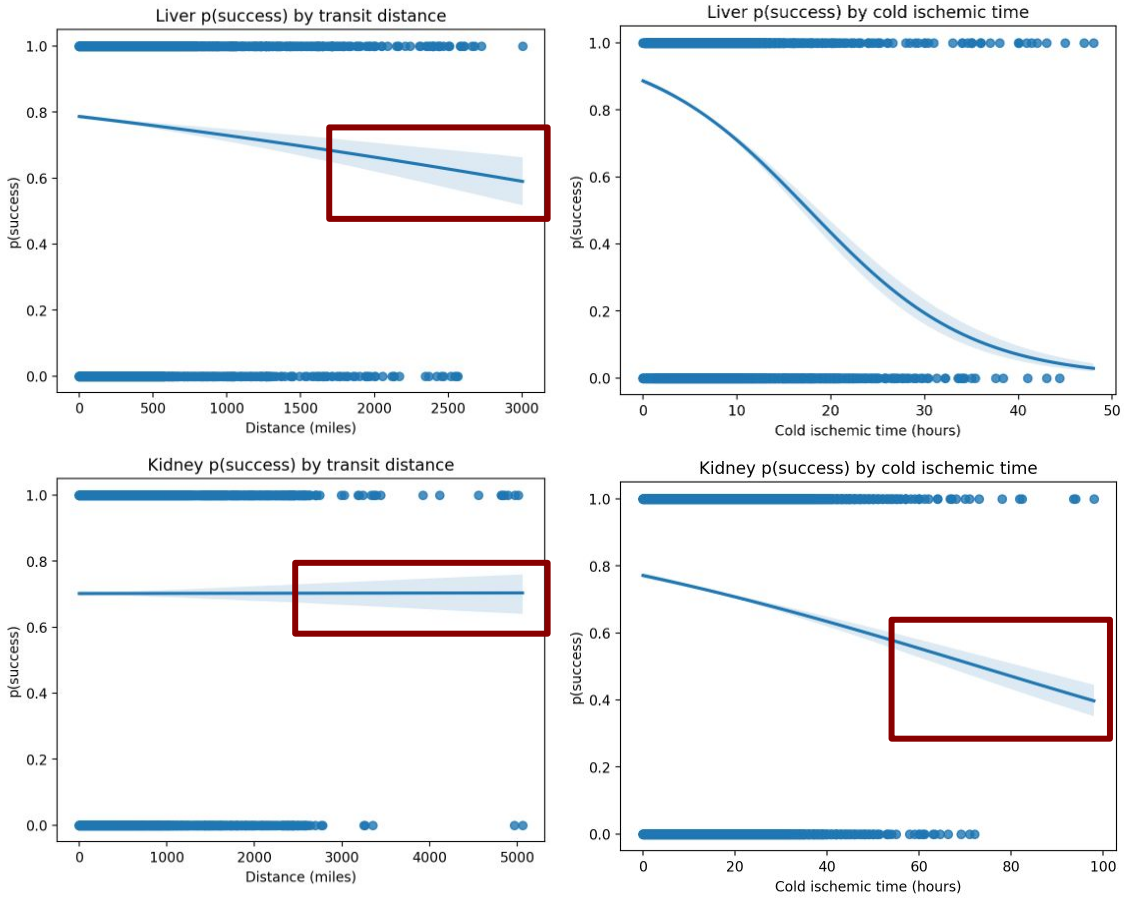



*We then use Maximum Likelihood Estimation (MLE) to find coefficients that make the observed data (the 0s and 1s) most likely to have happened.*

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

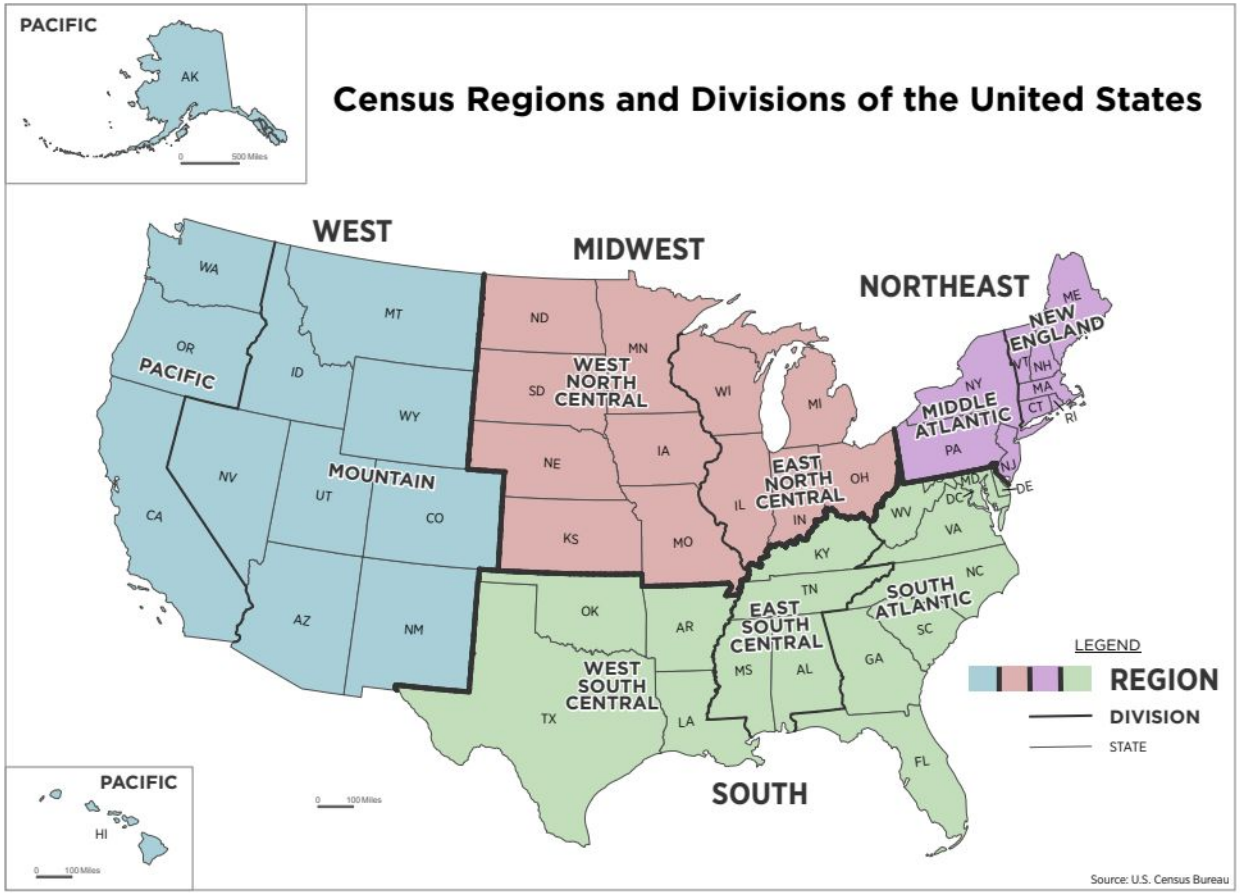
*Think back to calculus and limits – as defined p must stay in the range (0,1)*

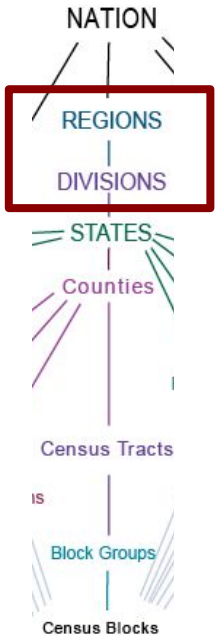
# Discussion – Confidence Bands




*Why does the width of the band increase as cold ischemic time rises?*

# Census Regions & Divisions




https://www2.census.gov/geo/pdfs/maps-data/maps/reference/us_regdiv.pdf